DDN AI400X3 24x7TiB nvme 4xNDR400 - 1 client

MLPerf Storage v2.0 Submission

System Name: AI400X3 24x7TiB nvme 4xNDR400 1client

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Benchmark Version: v2.0

Solution Type: Proprietary - EXAScaler (parallel filesystem)

Submission Division: CLOSED

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1 Hardware

The hardware is composed of 1 client and one server connected to the same switch.



1.1 Client

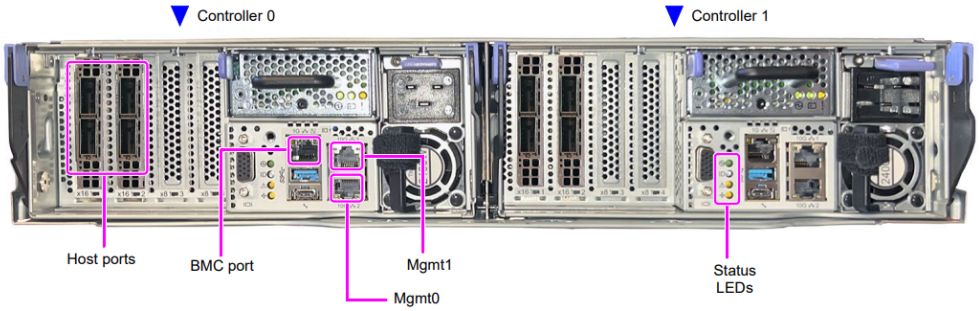
The client is a **bare-metal** Supermicro node (compute) with 2 CPU sockets and 478GiB of RAM based on Supermicro G1SMH motherboard (ARS-221GL-NR-01). The client is configured the same (hw & sw). The client communicates with the server through IB using one dedicated link.

- 1x NDR400 link total

1.2 Server

The server is **a commercial DDN AI400X3** appliance (storage) providing a single EXAScaler filesystem to the clients (one filesystem total). The configuration is the default one with 24x Phison NVME drives with 7TiB per drive. The server communicates with the clients through IB.

- The AI400X3 is a 2U rack-mounted appliance.
- 4x single-port NDR400 adapters for a total of 4x NDR400 links (one link per HCA). The following diagram shows two ports because it is the NDR200 flavor.



1.2.1 Power Requirements Table

The AI400X3 is designed with **redundant, active/active power supply units (PSUs)** to ensure continuous operation. It utilizes **two 2400W PSUs**, both of which supply power simultaneously to share the load and provide redundancy. This active/active configuration ensures that if one PSU fails, the remaining unit can fully support the system's **2400W**

design power requirement without interruption. This design eliminates single points of failure, as both PSUs remain active during normal operation and seamlessly take over in case of a failure. The PSUs are also hot-swappable, enabling maintenance without shutting down the system, ensuring high availability and reliability.

System component	Power supply unit	Nameplate rated power	Design power
DDN AI400X3	Power supply 1	2400 watts	2400 watts
	Power supply 2	2400 watts	
Totals		**4800 watts**	**2400 watts**

1.3 Switch

Mellanox QM8700 switches. The server and clients are connected to it.

2 Software

2.1 Client

The client's OS is Ubuntu 22.04.5 LTS (Jammy Jellyfish). The Mellanox OFED version is `OFED-internal`-25.04-0.6.1. The EXAScaler driver used on the client is `2.14.0_ddn212_8_ge87e912`. This is the GA version of the product with no modification applied.

2.2 Server

The AI400X3 runs EXAScaler SFA Rocky 6.3.3-2025032600 on top of the SFAOS stack (SFAOS version 12.7.0). The EXAScaler driver used on the server is `2.14.0_ddn198`. This is the GA version of the product with no modification applied. The OFED version is OFED-internal-24.10-1.1.4.

2.3 Switch

NA

3 Settings

3.1 Client

The standard EXAScaler 6.3.3 driver (GA) is installed on the client using the default installation procedure. The driver is used to access the remote EXAScaler filesystem.

3.1.1 EXAScaler driver configuration

The EXAScaler configuration file has been generated automatically following the default installation procedure. This configuration file is necessary to mount the filesystem. Content of /etc/modprobe.d/lustre.conf is:

```
# This file has been generated by exa-client-deploy
#
# Do not edit unless exa-client-deploy service is stopped & disabled
# e.g: 'systemctl status exa-client-deploy'
#

options lnet networks="o2ib(ibP18p1s0f0)"
options lnet lnet_transaction_timeout=100
options lnet lnet_retry_count=2
options ko2iblnd peer_credits=32
options ko2iblnd peer_credits_hiw=16
options ko2iblnd concurrent_sends=64
options ksocklnd conns_per_peer=0
```

As optimizations and customizations to the storage system are allowed in CLOSED Division, the following tunings were applied to each client:

```
# llama3-8B (checkpointing)
lctl set_param osc.*.checksums=0
# Cosmoflow & Resnet50 & Unet3D (training)
lctl set_param osc.*.max_pages_per_rpc=4M \
               osc.*.max_rpcs_in_flight=32 \
               osc.*.checksums=0 \
               llite.*.max_read_ahead_mb=0
```

3.2 Server

No tuning was applied on the DDN AI400X3 after the initial installation. The initial installation was automated using the latest version of our manufacturing script. The filesystem was mounted on the client after the script completed.

3.3 Switch

No tuning applied

4 Misc

4.1 Drop caches

The cache of both the client and server were dropped between iterations using the following function. Since 'clush -S' was used, both commands did work.

```
clear_caches() {
    # Clear cache on client
    clush -abS "sync; echo 3 > /proc/sys/vm/drop_caches"  || return 1
    # Clear cache on EXAScaler VMs
    ssh root@${exafsip}  "clush -abS 'sync; echo 3 > /proc/sys/vm/drop_caches'" \
        || return 1
}
```

4.2 Script

The run script uses the following bash options to ensure that all commands worked.

```
#!/bin/bash

set -u
set -e
set -x
set -o pipefail
```